BSSE FINAL PROJECT
<Requirements Specification>

VIRQA: Voice Based Real-Time Questioning System

Project Advisor

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Faculty of Information Technology & Computer Science

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University of Central Punjab

Software Requirements Specification

Version # 1

VIRQA: Voice Based Real-Time Questioning System

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Revision History

Name	Date	Reason For Changes	Version

Previous Phases Feedback

Idea Defence Feedback (Screenshot)

F25 Idea Defence Feedback + Proposal Submission Deadline

PO

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Cc Dr Adnan Ghafoor; Muhammad Tayyab Mir

9/1/2025

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Sr. #	SE-48
Project Title	VIRQA: VOICE BASED INTELLIGENT REAL-TIME QUESTIONING SYSTEM

Similar products already exist. It should not only be research based but development aspect should also be focused.

Add-ones:

- Candidate profile generation
- Employer dashboard
- Report generation
- BERT or Chat GPT 03 mini open source model
- Text conversion show on screen
- Domain specific

All suggested plus end product agreed.

1 Abstract

VIRQA is a voice-only AI interview platform that leverages large language models (LLMs), including BERT and open-source GPT alternatives, to deliver adaptive and skill-based interview experiences. It dynamically generates questions tailored to a candidate's competencies and evaluates spoken responses using advanced speech analysis. The system assesses tone, fluency, pauses, and hesitation patterns to infer confidence and communication skills. It then produces comprehensive performance reports, competency scores, and candidate profiles, helping organizations make data-driven hiring decisions. With its employer dashboard, automated evaluation, and scalable architecture, VIRQA offers an efficient, accessible, and consistent solution for recruitment, training, and talent development.

2 Introduction and Background

2.1 Product (Problem Statement)

Existing interview systems use static questions and text-based inputs, lacking real-time voice analysis, adaptive question generation, and semantic evaluation. They also fail to provide candidate profiles, employer monitoring tools, detailed reports, or real-time transcription. VIRQA addresses these gaps by developing a voice-based pipeline with domain-specific customization answer evaluation, and integrated features for profiling, dashboards, reports.

2.2 Background

The domain involves AI-driven HR technology, focusing on automated interviewing systems. Traditional platforms are limited to rule-based evaluations and static content, which do not adapt to candidate responses or analyze voice inputs in real-time. VIRQA builds on *LLMs*, and speech processing to create a more dynamic and fair system for recruitment.

2.3 Scope

The project covers design and development of a full-stack AI interview platform capable of:

- Generating domain-specific questions via *LLMs*
- Evaluating answers by tone, fluency, and semantics
- Producing reports and dashboards
- Enabling real-time voice transcription

Excludes external hardware and focuses on software development and evaluation accuracy.

2.4 Objective(s)/Aim(s)/Target(s)

The Objectives includes:

- Develop a text-based question generation module using *LLMs* for domain-specific technical, behavioral, and situational questions.
- Implement multi-metric answer evaluation (semantic accuracy, fluency, tone, completeness).
- Generate candidate profiles based on performance, skills, and confidence.
- Create an employer dashboard for real-time progress and score monitoring.
- Produce comprehensive performance reports with improvement recommendations.
- Enable real-time text transcription of spoken answers for accessibility.
- Compare AI and human evaluation scores.

2.5 Output candidate-specific scorecards and personalized feedback. Challenges

The challenges include:

- Designing domain-specific prompts for accurate question generation.
- Implementing semantic + fluency-based evaluation of spoken answers.
- Real-time transcription and on-screen display of text.

- Building integrated modules: candidate profiles, dashboards, and reports.
- Ensuring system scalability and handling *API*/*LLM* limitations.

2.6 Learning Outcomes

Upon completion of this project, we will be able to understand:

- Master Prompt Engineering: Design and optimize domain-specific prompts for *LLM*s like BERT or ChatGPT-3 Mini.
- Gain expertise in Speech and NLP: Analyze spoken answers using metrics like semantic accuracy, fluency, and tone.
- Develop skills in *LLM* integration: Work with *APIs* (e.g., Hugging Face, Whisper) for question generation and transcription.
- Learn multi-metric evaluation: Assess answers for relevance, completeness, and confidence.
- Create an end-to-end pipeline: Integrate voice-based question generation, evaluation, profiling, dashboards, reports, and transcription.

2.7 Nature of End Product

The final deliverable is a complete software system a voice-based AI interview platform with integrated *LLM*-driven question generation, evaluation, profiling, dashboards, reporting, and transcription.

2.8 Completeness Criteria

The table below shows the criteria:

Sr.	Criteria	Weightage %
1	Voice-based Question Generator	20%
2	Answer Evaluation Engine	20%
3	Candidate Profile Generation	15%
4	Employer Dashboard	15%
5	Report Generation	15%
6	Real-time Text Conversion Display	10%
7	Domain-specific Customization	5%

2.9 Business Goals

The business rules include:

- Automate candidate screening and evaluation
- Improve fairness and consistency in hiring
- Reduce recruiter workload
- Enable scalable, cost-effective recruitment and training
- Introduce data-driven insights for HR decisions

2.10 Related Work/ Literature Survey/ Literature Review

Several commercial and academic platforms have explored automated interviewing, but most rely on static question banks and rule-based evaluation **[4],[5].** This limits adaptability, fairness, and scalability in real-world applications.

Feature	Existing Solution	Our Proposed System
Voice-based Question Generation	Not Available	*LLM*-based, adaptive
Semantic Answer Evaluation	Keyword-based	Semantic accuracy, fluency
Candidate Profile Generation	Not Addressed	Comprehensive profiles
Employer Dashboard	Limited or Absent	Real-time monitoring
Report Generation	Generic or Absent	Detailed performance reports

2.11 Document Conventions

This SRS document adheres to standard **IEEE** conventions. *Italicized terms* denote external systems or components (e.g., *LLM APIs*, *AI*, *Operating System*). **Bold text** is used to emphasize key requirements. The document follows standard font guidelines:

- **Headings:** Times New Roman, 18pt, Bold
- **Subheadings:** Times New Roman, 14pt, Bold
- **Body Text:** Times New Roman, 12pt, Regular

3 Overall Description

3.1 Product Features

The below are features:

- Adaptive voice-based question generation using LLMs.
- Multi-metric evaluation of spoken answers (semantic, fluency, tone, completeness).
- Candidate profile creation summarizing skills and performance.
- Employer dashboard for real-time monitoring.
- Automated performance reports with recommendations.
- Real-time speech-to-text transcription display.

3.2 Domain-specific customization for job roles. User Classes and Characteristics

The customization of roles includes:

- **Candidates**: Job applicants or trainees; frequent users; require intuitive voice interface; varying technical expertise.
- **Employers/HR Managers**: Monitor interviews; use dashboards and reports; high priority users; technical expertise assumed; focus on analytics.
- **Administrators**: System setup and customization; infrequent users; high privilege level.

3.3 Operating Environment

The environment most suitable is:

- **Hardware**: Standard desktop/laptop with microphone (no special requirements).
- **OS**: Windows/Linux/Mac (cross-platform).
- **Software**: Python 3.x runtime; compatible with web browsers for UI (e.g., Chrome).
- **API:** Must coexist with APIs like Hugging Face, Whisper or any other suitable *API*.

3.4 Design and Implementation Constraints

- **Models**: Use open-source LLMs API (e.g., BERT via Hugging Face) or any other to avoid high costs; fallback from paid APIs if needed.
- **Language**: Python for **AI**; **MERN** for application development.

- **Security**: Adhere to data privacy standards (no storage of sensitive audio without consent).
- **Tools**: VS Code, GitHub for development; no proprietary hardware.
- **Constraints**: *API* rate limits; ensure offline-capable components where possible.

3.5 Assumptions and Dependencies

The possible assumptions and dependency of the system includes:

- Stable Internet connection
- Availability of open-source LLMs and speech models
- Continuous advisor feedback and GitHub access
- Stable versions of software using

4 Functional Requirements

4.1 Use-Case Diagram

4.2 Add New Candidates

Identifier	UC-1	
Purpose	Allow new candidate to create accounts.	
Priority	High	
Pre-conditions	Must Apply for any selection	
Post-conditions	Account created and saved in database.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate Apply for any position	N/A
2	Employee enter Employee details	System checks the possibilities.
3	Employee create account	User record saved in database
4	Candidate waits for OTP	System sends OTP to email
5	Candidate verifies account	System displays account created
Alternate Course of Action		
S#	**Actor Action**	**System Response**
4a	Candidate submits form	System displays "Email already registered"

Table 1: UC-1

4.3 Candidate Login

Identifier	UC-2	
Purpose	Authenticate candidate for login access.	
Priority	High	
Pre-conditions	Candidate account exists.	
Post-conditions	Candidate successfully logged in	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate opens login page	System displays login form.
2	Candidate enters credentials	System validates fields
3	Credentials are valid	System verifies email and password.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3a	Invalid credentials	System displays "Incorrect email or password"

Table 2: UC-2

4.4 Candidate Update Profile

Identifier	UC-3	
Purpose	Allow candidate to update profile details.	
Priority	Medium	
Pre-conditions	Candidate logged in	
Post-conditions	Updated details stored in database	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate opens profile page	System fetches existing data
2	Candidate edits skills, description, and image	System validates inputs
3	Candidate saves profile	Updated info!
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Invalid format	System displays “Invalid data input.”

Table 3: UC-3

4.5 Candidate Join Interview

Identifier	UC-4	
Purpose	Allow candidate to join scheduled interview	
Priority	High	
Pre-conditions	Interview scheduled by employer	
Post-conditions	Candidate connected to interview session	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate enters join password	System validates credentials
2	Password valid	System allows access to interview session
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Invalid join password	System displays “Invalid interview id”

Table 3: UC-3

4.6 Record Candidate Voice

Identifier	UC-5	
Purpose	Enable candidate to record voice answers	
Priority	High	
Pre-conditions	Interview started	
Post-conditions	Answer saved as voice file in database	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate starts recording	System starts recording
2	Candidate speaks answer	System captures audio
3	Candidate exits after interview	Interview conducted successfully!
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1a	Microphone blocked	System displays “Please allow microphone access.”

Table 5: UC-5

4.7 Candidate Result

Identifier	UC-6	
Purpose	View results and feedback after interview	
Priority	Medium	
Pre-conditions	Interview evaluated by AI interviewer	
Post-conditions	Candidate views score and suggestions	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate opens results	System displays score, confidence, and suggestions
2	Candidate reviews	System waits
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1a	Candidate opens results	System displays “Results not ready yet.”

Table 6: UC-6

4.8 Reset Password

Identifier	UC-7	
Purpose	Reset forgotten password	
Priority	Low	
Pre-conditions	Candidate email exists.	
Post-conditions	Password successfully changed.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate clicks forgot password	System requests email.
2	Candidate enters email	System sends OTP link
3	Candidate resets password	System updates record
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Email not found	System displays “Account not found.”

Table 7: UC-7

4.9 Logout Candidate

Identifier	UC-8	
Purpose	Securely logout candidate.	
Priority	Low	
Pre-conditions	Candidate logged in	
Post-conditions	Tokens cleared, session closed.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate clicks logout	System deletes tokens
2	Candidate views successfully logout	System redirects to login page
3	Candidate resets password	System updates record
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1a	Timeout reached	Auto logout executed

Table 8: UC-8

4.10 Register Institute

Identifier	UC-9	
Purpose	Register institute account	
Priority	High	
Pre-conditions	Institute email unique	
Post-conditions	Institute Registered successfully	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Institute fills registration form	System validates inputs
2	Record created	System sends OTP link
3	Institute verifies OTP	System updates record
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Email already used	System display "Email already registered."

Table 9: UC-9

4.11 Add New Employees

Identifier	UC-10	
Purpose	Add new employee or interviewer	
Priority	High	
Pre-conditions	Institute logged in	
Post-conditions	Employee account created	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Institute opens add employee	System displays form
2	Details entered	Record saved in database
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1a	Email exists	System shows "Employee already exists."

Table 10: UC-10

4.12 Employee Login

Identifier	UC-11	
Purpose	Allow employee/interviewer to log in	
Priority	High	
Pre-conditions	Employee registered	
Post-conditions	Access granted to dashboard	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Employee enters credentials	System verifies
2	Employee sees successfully logged in	System directs to dashboard
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Wrong password	System displays “Invalid Credentials”

Table 11: UC-11

4.13 Interview Create

Identifier	UC-12	
Purpose	Create new interview	
Priority	High	
Pre-conditions	Employee logged in	
Post-conditions	Interview Successfully created	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Employer enters title, description and other required details	System validates
2	Employee clicks confirm	System creates interview
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Employee clicks cancel	System cancels interview creation

Table 12: UC-12

4.14 Interview Question Generation

Identifier	UC-13	
Purpose	Generate adaptive interview questions using AI	
Priority	High	
Pre-conditions	Interview started and candidate is in interview	
Post-conditions	Questions generated and stored in system	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate starts interview	System generates questions
2	Candidate answers questions	System evaluates and generate adaptive question
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Candidate leaves interview	System stops generating question

Table 13: UC-13

4.15 Evaluate Candidate

Identifier	UC-14	
Purpose	Automatically evaluate candidate responses	
Priority	High	
Pre-conditions	Candidate submit answer and voice is also working correctly	
Post-conditions	Scores calculated and stored	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate answers question	System evaluates answers and frequency
2	Candidate continues answering	System adapts on basis of previous answers
2	Candidate complete interview	System calculates total score on basis of formula applied

Table 14: UC-14

4.16 Feedback Report Generate

Identifier	UC-15	
Purpose	Generate feedback for candidate based on evaluation	
Priority	High	
Pre-conditions	Evaluation completed	
Post-conditions	Feedback report generated	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Employer clicks generate feedback	System analyzes performance data
2	Candidate complete interview	System calculates total score and generate report

Table 15: UC-15

4.17 View Candidate Results

Identifier	UC-16	
Purpose	Allow employer to view results of all candidates	
Priority	Medium	
Pre-conditions	Completed interviews available	
Post-conditions	Results displayed on dashboard	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Employer opens results tab	System retrieves all interview results
2	Employer views performance by filters	System displays detailed metrics
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Employer opens dashboard with no data	System shows “No interviews completed yet.”

Table 16: UC-16

4.18 Admin Login

Identifier	UC-17	
Purpose	Allow administrator to log into system	
Priority	High	
Pre-conditions	Admin account exists	
Post-conditions	Admin dashboard displayed	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Admin enters credentials	System verifies login data
2	Admin clicks login	System grants access to dashboard
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Admin enters invalid password	System displays “Invalid credentials.”

Table 17: UC-17

4.19 Logout Administrator

Identifier	UC-18	
Purpose	Logout administrator securely	
Priority	Low	
Pre-conditions	Admin logged in	
Post-conditions	Session terminated	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Admin clicks logout	System ends session
2	Admin redirected to login page	System clears tokens
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Admin session expires	System auto-logs out admin

Table 18: UC-18

4.20 Real-Time Voice Question Playback

Identifier	UC-19	
Purpose	Deliver LLM-generated questions audibly to the candidate also	
Priority	High	
Pre-conditions	Question text generated successfully in backend	
Post-conditions	Audio question played clearly for the candidate	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate starts interview	System generates question audio
2	Candidate listens to question	System plays audio output.
3	Candidate prepares answer	System waits for response

Table 19: UC-19

4.21 Real-Time Answer Transcription

Identifier	UC-20	
Purpose	Transcribe candidate's spoken answers into text for accessibility and evaluation	
Priority	High	
Pre-conditions	Candidate microphone access granted	
Post-conditions	Transcription displayed and stored	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate speaks answer	System records audio
2	Candidate completes response	System sends audio to transcription API
3	Candidate waits for next question	System uses this for evaluation

Table 20: UC-20

4.22 Adaptive Question Difficulty

Identifier	UC-21
Purpose	Automatically adjust the difficulty of upcoming questions based on candidate performance
Priority	High
Pre-conditions	Candidate has answered at least one question
Post-conditions	System selects next question with updated difficulty level
Typical Course of Action	

S#	Actor Action	System Response
1	Candidate answers question	System evaluates accuracy and response time
2	Candidate proceeds to next question	System selects higher or lower difficulty accordingly

Table 21: UC-21

4.23 Real-Time Voice Sentiment Analysis

Identifier	UC-22	
Purpose	Detect candidate's stress level and confidence from voice tone	
Priority	High	
Pre-conditions	Microphone access granted by candidate	
Post-conditions	A part of score successfully generated from voice	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate speaks answer	System analyzes tone and pitch in real time
2	Candidate completes sentence	System logs emotion and confidence score

Table 22: UC-22

4.24 Topic Coverage Evaluation

Identifier	UC-23	
Purpose	Assess whether the candidate addressed all key areas of the question	
Priority	High	
Pre-conditions	Clean transcript available	
Post-conditions	Topic coverage score computed	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Candidate finishes response	System verifies answer by using semantic evaluation
2	Employer reviews report	System displays suggestions regarding to answer if there
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Candidate gives invalid answer	System gives possible answer for the question

Table 23: UC-23

4.25 Interview Replay and Review

Identifier	UC-24	
Purpose	Allow employer to replay candidate's recorded interview for verification	
Priority	Medium	
Pre-conditions	Interview recording stored successfully	
Post-conditions	Playback interface displayed	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Employer selects candidate interview	System loads audio and transcript
2	Employer plays recording	System plays audio
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Recording missing	System shows "Audio not available."

Table 24: UC-24

4.26 Remove Employer/interviewer

Identifier	UC-25	
Purpose	Allow admin (institute) to remove an existing employee from the system	
Priority	Low	
Pre-conditions	Employer logged in; employee record exists	
Post-conditions	Employee account removed or deactivated successfully	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Admin opens manage employees section	System displays all registered employees
2	Employer selects an employee to remove	System confirms removal action
3	Employer clicks remove	System deletes or deactivates the selected employee record
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3a	Employer cancels the action	System retains employee record and shows "Operation cancelled."

Table 25: UC-25

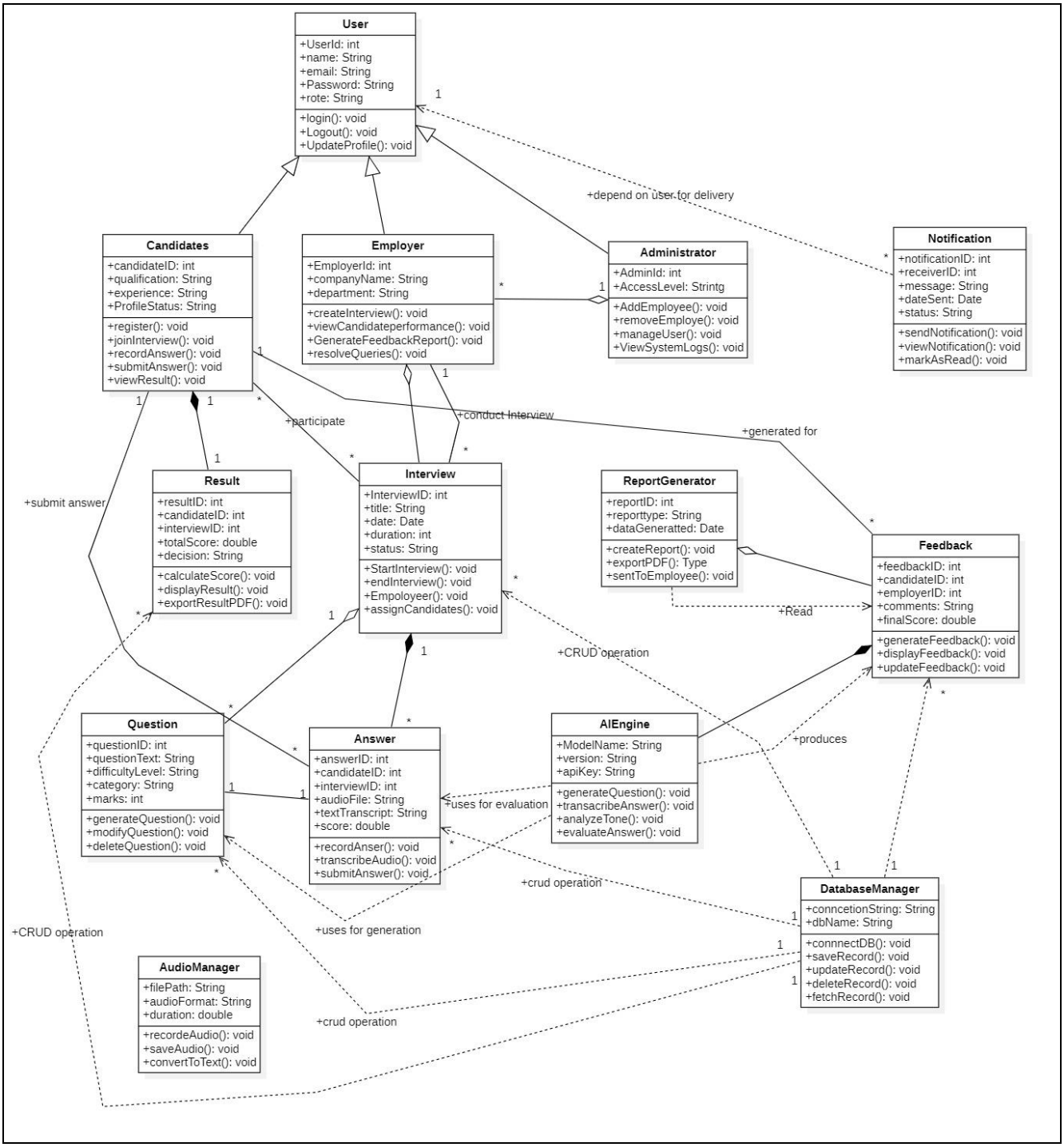
4.27 Update Employee Profile

Identifier	UC-26	
Purpose	Allow employee to update personal profile details such as name, description, or image	
Priority	Low	
Pre-conditions	Employee logged in	
Post-conditions	Updated employee information saved in the database	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Employee opens “My Profile” page	System displays existing employee data
2	Employee edits description, profile picture, or email	System validates inputs
3	Employer clicks remove	System updates record
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Employee provides invalid data format	System displays “Invalid input, please recheck.

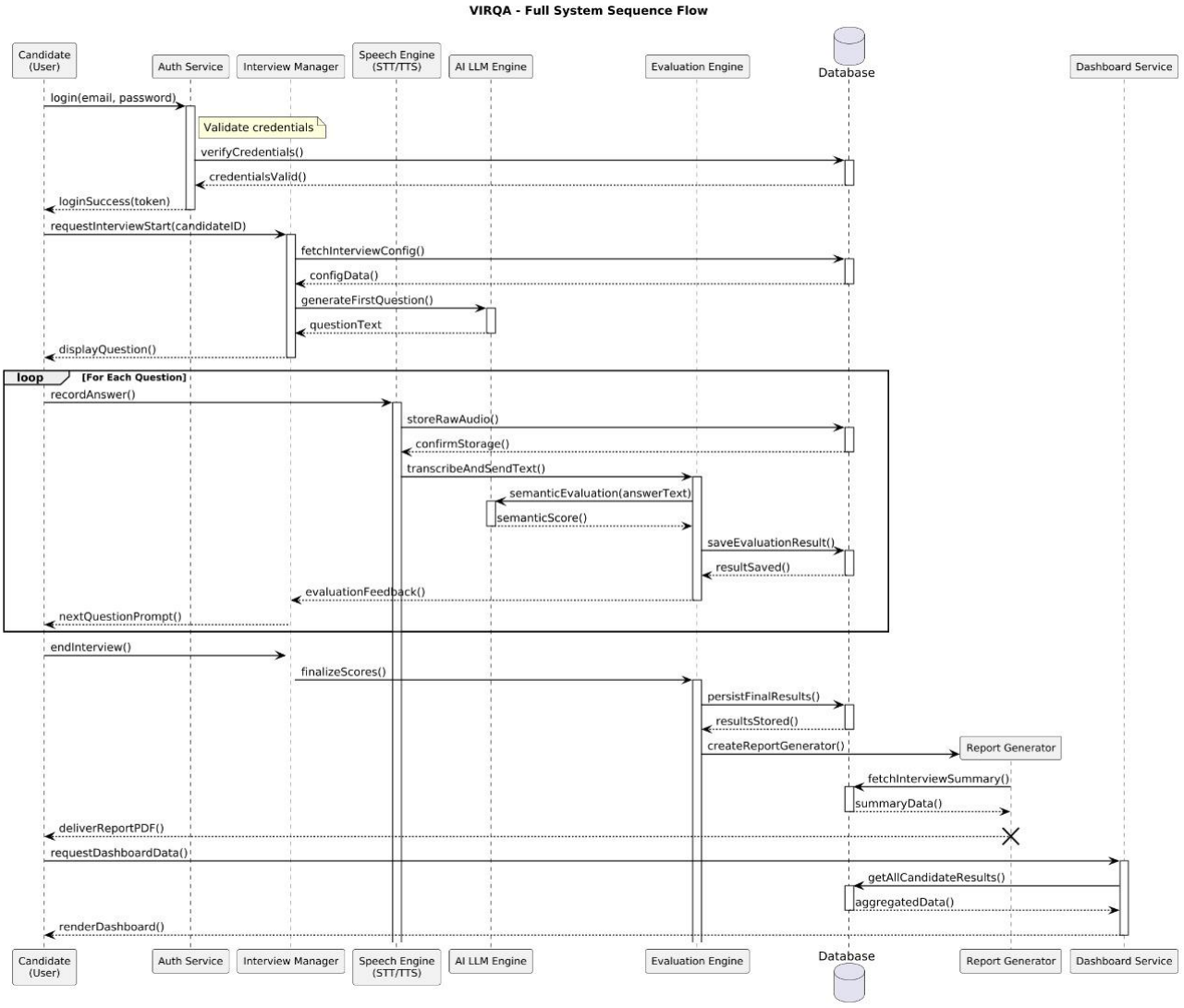
Table 26: UC-26

4.28 Requirements Analysis and Modeling

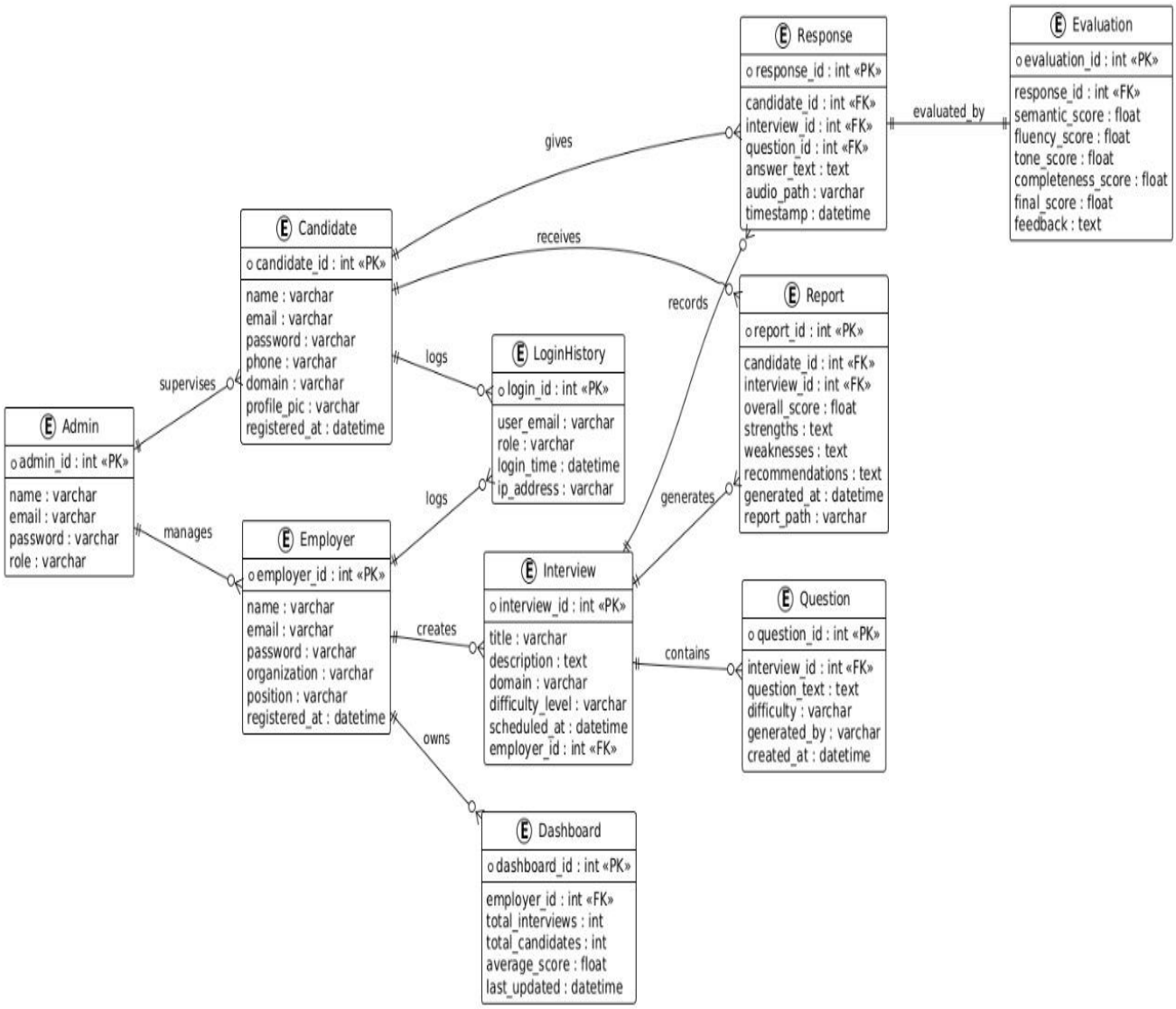
4.28.1 Class Diagram



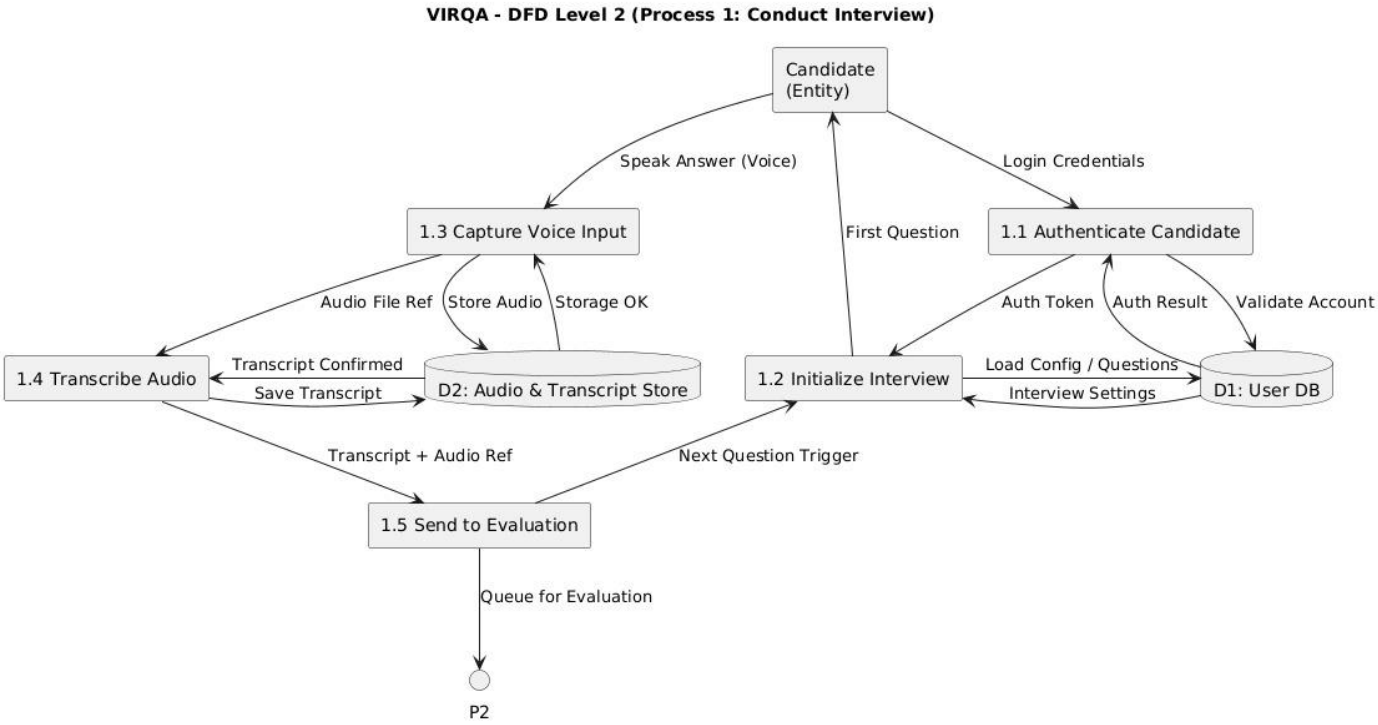
4.28.2 Sequence Diagram



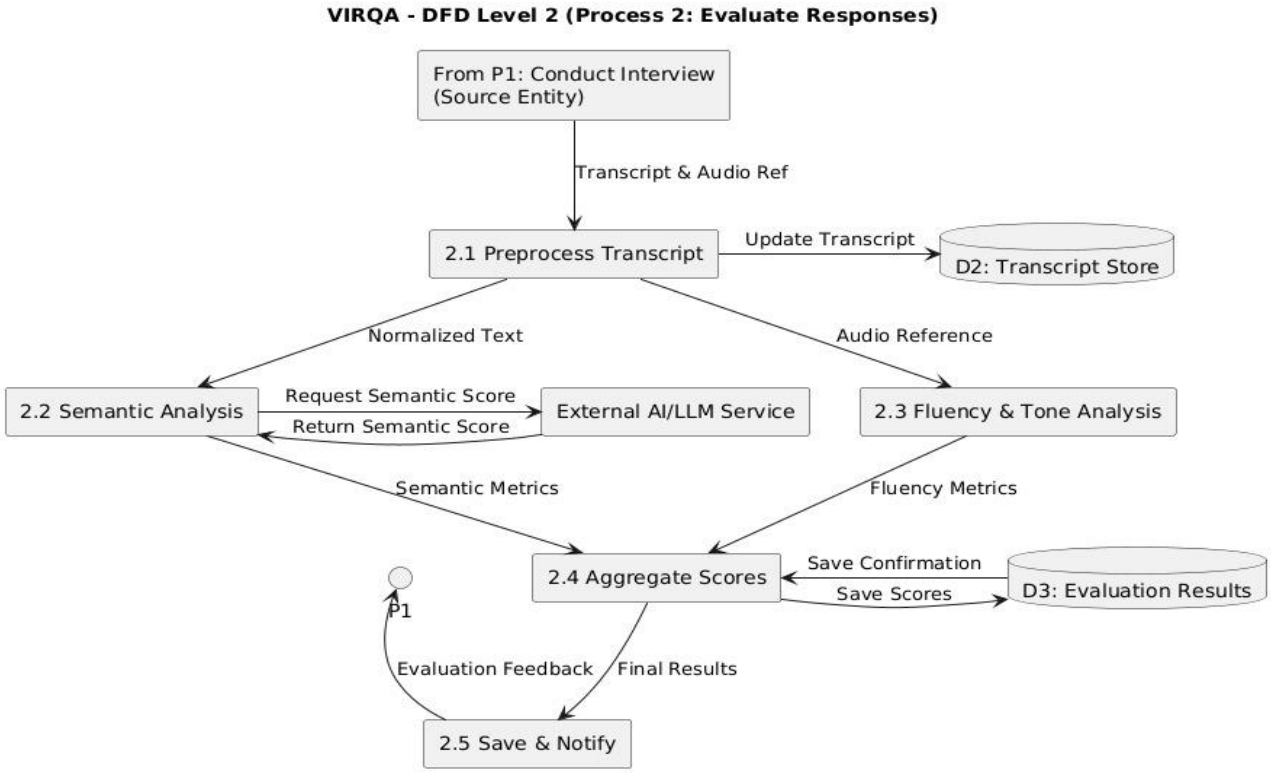
4.28.3 Data Flow Diagram



4.28.4 Data Flow Diagram L1 (Process 1)



4.28.5 Data Flow Diagram L1 (Process 2)



5 Nonfunctional Requirements

5.1 Performance Requirements

The performance requirements are:

- The system shall generate and play a domain-specific question using the LLM within **30 seconds** after the employer initiates question generation.
- The system shall transcribe the candidate's spoken response into text within **20 seconds** after the candidate finishes speaking.
- The system shall evaluate a candidate's answer across semantic, fluency, tone, completeness metrics within 30 **seconds** of submission.
- The system shall generate a complete candidate performance profile within **45 seconds** after interview completion.
- The system shall display updated interview progress and candidate metrics on the employer dashboard every **5 seconds** during an ongoing session.
- The system shall generate personalized feedback and suggestions for the candidate within **8 seconds** after evaluation is completed.
- The system shall adjust the next question's difficulty level within **8 seconds** after receiving the previous answer's score.
- The system shall maintain a minimum transcription accuracy of **80%** for English-language responses in standard audio conditions.
- The system shall ensure that audio playback starts within **10 seconds** after question text generation for smooth voice-based delivery.
- The system shall complete report generation and display the downloadable performance summary within **5 seconds** after the employer requests it.
- The system shall store and retrieve any candidate's interview history within **4 seconds** from the database query request.
- The system shall process up to **10 concurrent interviews simultaneously** without noticeable performance degradation.
- The system shall automatically reconnect to the server within **10 seconds** after network restoration during a live interview.
- The system shall ensure **80% reliability** in capturing and saving all candidate audio responses during interviews.
- The system shall complete candidate and employer dashboard loading within **10 seconds** on a stable 5 Mbps connection.

- The system shall initiate text-to-speech question playback within **10 seconds** of question generation to maintain conversational flow.
- The system shall ensure that at least **85% of LLM and evaluation API requests succeed** on the first attempt.
- The system shall download and export performance reports in PDF format within **12 seconds** of the user request.
- The system shall complete audio file uploads to the server within **10 seconds** for standard 10 MB files.
- The system shall maintain an average **server response time of less than 1.5 seconds** for all database read/write operations.
- The system shall ensure an overall **95% uptime** for all user-facing services during operational hours.

5.2 Safety Requirements

The safety requirements are:

- The system shall prevent unauthorized users from accessing candidate, employee, or institute data through secure authentication and token management.
- The system shall automatically encrypt all sensitive data (passwords, tokens, voice recordings) to prevent data leakage or misuse.
- The system shall ensure that all recorded audio and generated transcripts are stored securely and deleted permanently upon account removal.
- The system shall prevent AI models from generating or displaying inappropriate, biased, or offensive content by applying pre-defined moderation filters.
- The system shall restrict the use of recorded candidate voice data strictly for evaluation and report generation purposes, preventing any external distribution or re-use.
- The system shall immediately stop recording and notify the user if microphone access permissions are revoked during an interview.
- The system shall ensure that no sensitive information (e.g., API keys, passwords, internal tokens) is ever displayed to end-users or stored in client-side code.
- The system shall provide a safe exit mechanism allowing candidates or interviewers to end the session safely without corrupting interview data.
- The system shall prevent simultaneous multiple logins using the same credentials to avoid session hijacking or data inconsistency.
- The system shall restrict AI access to only approved APIs and models that comply with organizational and academic data safety standards.

- The system shall prevent unverified or expired access tokens from being used for any request, ensuring safe session control.
- The system shall ensure that all AI-generated content follows institutional and ethical guidelines, minimizing psychological discomfort or offensive outputs.
- The system shall log all exceptions and runtime errors with timestamps for safe post-incident debugging without exposing user data.
- The system shall comply with safety and privacy standards equivalent to **ISO/IEC 27001** (Information Security Management) and **GDPR-like principles** for data protection **[8],[9].**

5.3 Security Requirements

The security requirements include:

- The system shall require all users (candidates, employees, and administrators) to authenticate using a valid email and password before accessing any protected feature.
- The system shall encrypt all user passwords using a strong hashing algorithm before storing them in the database.
- The system shall issue unique access and refresh tokens upon successful authentication, which automatically expire after **60 minutes** of inactivity.
- The system shall implement role-based access control (RBAC) ensuring each user type (Candidate, Employee, Admin) can only access permitted modules.
- The system shall immediately invalidate all active tokens when a user logs out or the session expires.
- The system shall perform input validation and sanitization on all user-provided data to prevent SQL injection and XSS attacks.
- The system shall ensure that API keys and credentials are stored securely on the server side and never exposed to frontend code.
- The system shall restrict direct access to the database to authorized backend processes only.
- The system shall reject any API request lacking a valid and non-expired access token.
- The system shall restrict uploaded file types to .pdf formats only to avoid malicious uploads.
- The system shall prevent concurrent logins from multiple devices using the same credentials.
- The system shall use secure cookies with the “HttpOnly” and “Secure” attributes enabled for all session management.
- The system shall prevent public access to internal server files, configuration directories, or logs.
- The system shall implement API rate limiting to mitigate denial-of-service.

5.4 Additional Software Quality Attributes

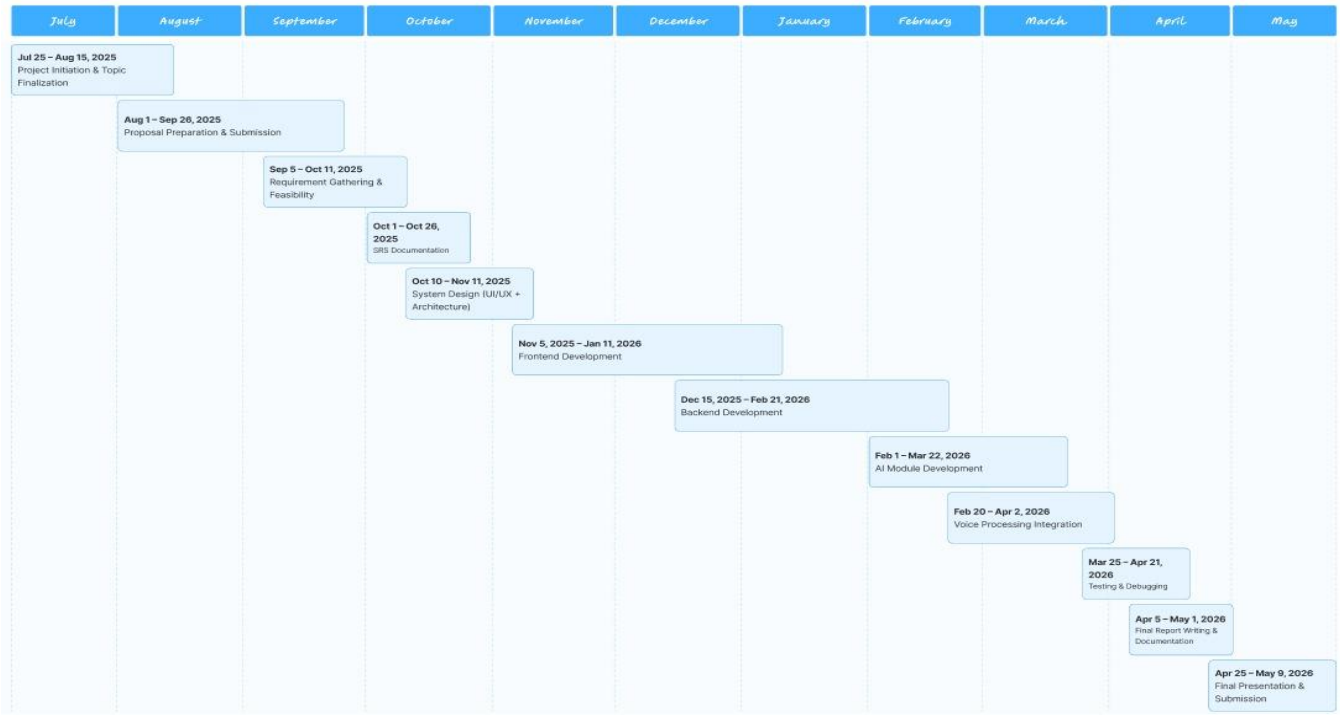
The additional attributes include usability and other requirements that are follow:

- The system shall allow a new candidate to register, set up a profile, and start an interview session within **5 minutes** without external guidance.
- The interface shall display all major functions (Dashboard, Interview, and Results) within 3 clicks from the home page.
- The system shall ensure that all navigation buttons, input fields, and icons are labeled clearly and consistently throughout the interface.
- The system shall operate on Windows, and Linux with no more than 10% configuration differences**.**
- The system shall maintain **95% uptime** during official operating hours.
- The system shall recover automatically from temporary failures within **10 seconds** after network restoration.
- The system shall allow administrators to add new scoring metrics or question categories without changing the database schema.
- The system shall support configuration-based time limits and difficulty scaling for interviews.
- The system shall integrate with at least **two third-party AI APIs** (e.g., OpenAI, Google Speech) without major architectural changes.
- The system shall communicate between frontend, backend, and AI modules using **standard RESTful APIs** with JSON format.
- The system shall maintain a **modularity index ≥ 0.75**, ensuring components can be updated independently.
- The entire system setup process on a new environment shall not exceed **30 minutes**.
- The AI question generation and evaluation modules shall be designed as independent reusable micro services that can be integrated into other platforms.
- Common utility functions (logging, validation, token management) shall be developed as shared components reused across all modules.
- The system shall handle up to 100 concurrent interview**s** with no more than **10% performance degradation**.
- The system shall gracefully handle unexpected runtime errors by displaying a user-friendly message and logging the incident automatically.

- The system shall maintain automated test scripts covering at least **80%** of backend functions**.**

6 Revised Project Plan

6.1 Gant Chart



6.2 Project Progress Summary

Till now, we have completed the first phase of our project. The topic "**VIRQA: Voice Based Real-Time Questioning System**" has been finalized and approved. The proposal, feasibility study, and complete SRS document have also been prepared and submitted on time.
The project is now moving to the **system design phase**, where we will design the user interface and overall system architecture. So far, the work is going as planned, and the project is **on schedule** with no major delays.

6.3 Update Project Plan Overview

Phase	Planned Duration	Key Deliverables	Status
Project Initiation & Topic Finalization	25 Jul – 1 Sep 2025	Approved topic, team formation	Completed
Proposal Preparation & Submission	1 Sep – 17 Sep 2025	Approved proposal & advisor feedback	Completed
SRS Documentation	17 Sep – 17 Nov 2025	Full SRS (Phase 1 submission)	Completed
System Design (UI/UX + Architecture)	17 Nov – 22 Dec 2025	Architecture diagram, design models	In Progress
Frontend Development	22 Dec 2025 – 10 Feb 2026	Candidate & Employer portals	In Progress
Backend Development	10 Feb – 01 Mar 2026	REST API, Database schema	In Progress
AI Module Development	01 Mar – 15 Mar 2026	Question generation, evaluation models	Upcoming
Voice Processing Integration	15 Mar – 25 Mar 2026	Speech-to-text & TTS integration	Upcoming
Testing & Debugging	25 Mar – 30 Mar 2026	Test report, fixes	Upcoming
Final Report & Presentation	30 Mar – 29 Apr 2026	Final documentation & viva	Upcoming

7 References

[1] I. Sommerville, *Software Engineering*, 10th Edition, Pearson Education, 2015.

[2] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th Edition, McGraw-Hill Education, 2020.

[3] IEEE Computer Society, *IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830-1998)*, 1998.

[4] OpenAI, "GPT Models Documentation," Available at: https://platform.openai.com/docs, Accessed: Nov. 11, 2025.

[5] Hugging Face, "Transformers: Machine Learning Model Hub," Available at: https://huggingface.co/docs, Accessed: Nov. 11, 2025.

[6] Google Cloud, "Speech-to-Text API Documentation," Available at: https://cloud.google.com/speech-to-text/docs, Accessed: Nov. 11, 2025.

[7] OWASP Foundation, "OWASP Top 10 – Security Risks for Web Applications," Available at: https://owasp.org/www-project-top-ten/, Accessed: Nov. 11, 2025.

[8] ISO/IEC 27001, *Information Security Management Systems – Requirements*, International Organization for Standardization, 2013.

[9] European Union, "General Data Protection Regulation (GDPR) 2016/679," Available at: https://gdpr.eu/, Accessed: Nov. 11, 2025.

Appendix A: Glossary

This glossary defines all acronyms and abbreviations used in the **VIRQA: Voice-Based Real-Time Questioning System** SRS document. Each term is defined at its first occurrence in the text.

Sr.	Acronym/Synonym	Full Form / Definition
1	**LLM**	Large Language Model
2	**AI**	Artificial Intelligence
3	**API**	Application Programming Interface
4	**OTP**	One-Time Password
5	**MFA**	Multi-Factor Authentication
6	**RBAC**	Role-Based Access Control
7	**UI**	User Interface
8	**UX**	User Experience
9	**TTS**	Text-to-Speech
10	**STT**	Speech-to-Text
11	**AES**	Advanced Encryption Standard
12	**TLS**	Transport Layer Security
13	**HTTP**	Hypertext Transfer Protocol
14	**HTTPS**	Hypertext Transfer Protocol Secure
15	**SQL**	Structured Query Language
16	**DoS**	Denial of Service
17	**WCAG**	Web Content Accessibility Guidelines
18	**JSON**	JavaScript Object Notation
19	**AWS**	Amazon Web Services
20	**OWASP**	Open Web Application Security Project
21	**GDPR**	General Data Protection Regulation
22	**ISO**	International Organization for Standardization
23	**IEC**	International Electrotechnical Commission

24	**URL**	Uniform Resource Locator
25	**IP**	Internet Protocol

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
…					

Table 2: List of non-trivial defects